

1)

$P_{1,2}$ = raíces del polinomio $z^2 - 0.3z^{-1} + 0.02$

$$P_{1,2} = \frac{0.3 \pm \sqrt{0.09 - 0.08}}{2} = \begin{cases} 0.2 \\ 0.1 \end{cases}$$

$$H_z(z) = T \cdot \left[\frac{A}{1 - 0.2z^{-1}} + \frac{B}{1 - 0.1z^{-1}} \right]$$

$$\begin{cases} A+B=1 \\ -0.1A-0.2B=0 \end{cases} \Rightarrow \begin{cases} A=2 \\ B=-1 \end{cases}$$

$$h(kT) = 2 \cdot (0.2)^k 1_0(kT) - (0.1)^k 1_0(kT)$$

$$E_h = T \cdot \sum_{k=-\infty}^{+\infty} |h(kT)|^2 = T \cdot \sum_{k=0}^{+\infty} (2 \cdot 0.2^k - 0.1^k)^2 =$$

$$= T \cdot 4 \sum_{k=0}^{+\infty} (0.04)^k + T \sum_{k=0}^{+\infty} (0.01)^k - 4T \sum_{k=0}^{+\infty} (0.02)^k$$

$$= \frac{4T}{1-0.04} - \frac{4T}{1-0.02} + \frac{T}{1-0.01}$$

$$\begin{aligned} 2) \quad x(t) &= 0.4 \cos\left(\pi t - \frac{\pi}{8}\right) & a(t) &= e^{-0.25(t-4)^2} \\ &= 0.4 \cos\left(2\pi \frac{1}{2} t - \frac{\pi}{8}\right) & &= e^{-\pi \left(\frac{t-4}{2\sqrt{\pi}}\right)^2} \\ H(f) &= 2\sqrt{\pi} e^{-\pi(2\sqrt{\pi}f)^2} e^{-j2\pi f \cdot 4} \end{aligned}$$

$$f_0 = \frac{1}{2}$$

$$y(t) = 0.4 |H(f_0)| \cos\left(\pi t - \frac{\pi}{8} + \angle H(f_0)\right)$$

$$H(f_0) = H\left(\frac{1}{2}\right) = 2\sqrt{\pi} e^{-\pi^2} e^{-j4\pi} = 2\sqrt{\pi} e^{-\pi^2}$$

$$|H(f_0)| = 2\sqrt{\pi} e^{-\pi^2} \angle H(f_0) = 0$$

$$y(t) = 2\sqrt{\pi} 0.4 e^{-\pi^2} \cos\left(\pi t - \frac{\pi}{8}\right)$$

3) $w(kT)$ è gaussiana e media nulla, in quanto
 $w(kT) = x(kT) - 0.4 x(kT-T)$.

La correlazione di $w(kT)$ è

$$E[w(kT+2T)w(kT)] = E[(x(kT+2T) - 0.4x(kT+2T-T)) \\ (x(kT) - 0.4x(kT-T))]$$

$$= r_x(kT) - 0.4r_x(kT+T) - 0.4r_x(kT-T) + 0.16r_x(kT)$$

$$r_w(kT) = 1.16r_x(kT) - 0.4r_x(kT+T) - 0.4r_x(kT-T) \quad (*)$$

$R_x(f)$ è la periodogramma di R_0 tra 0 e $2fT$, quindi

$$r_x(kT) = \frac{R_0}{2T} \text{sinc}^2\left(\frac{kT}{2T}\right) \quad (\text{compromesso})$$

$y = w(kT+T) + w(kT+2T)$ è gaussiana con media nulla e varianza

$$E[y^2] = E[w^2(kT+T) + w^2(kT+2T) + 2w(kT+T)w(kT+2T)] \\ = r_w(0) + r_w(0) + 2r_w(T) = \sigma_y^2$$

Sostituendo (*) e (**) si trova σ_y^2 .

$$P[y > 1] = Q\left(\frac{1}{\sigma_y}\right)$$

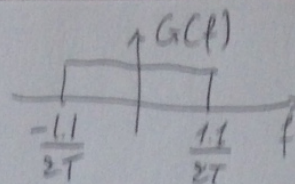
4) Dato che $f_x(e) = 0$ per $|e| > 1$, il fondo noise del quantizzatore viene posto per $e = 1$.

$$\Delta = \frac{2V}{L} = \frac{2}{L}. \text{ Ad altri bit vale } M_e = \frac{\Delta^2}{12} = \frac{1}{3L^2}$$

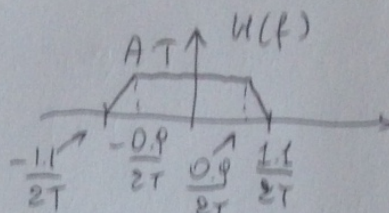
$$M_x = \int_{-1}^1 e^2 (1-e^2) \frac{3}{4} de = \frac{3}{4} \left(\frac{e^3}{3} - \frac{e^5}{5} \right) \Big|_{-1}^1 = \frac{1}{5}$$

$$\Lambda = \frac{M_x}{M_e} = \frac{3}{5} L^2$$

$$5) G(f) = V_0 \frac{T}{1.1} \text{rect}\left(\frac{fT}{1.1}\right)$$



$$L(f) = A_m$$



$$C(f) = G(f) \cdot L(f) H(f) = A_m \frac{V_0 T}{1.1} \cdot H(f)$$

$$c(t) = A_m \frac{V_0 T}{1.1} A \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{0.1t}{T}\right) \quad \text{вд Ньпуотр}$$

$$V_0' = C(0) = A_m A \frac{V_0 T}{1.1} = \frac{V_0 T}{1.1}$$

$$\sigma_n^2 = R_0 E_n = R_0 \int_{-\infty}^{+\infty} |H(f)|^2 df = R_0 \left(A_m^2 T^2 \cdot \frac{0.4}{T} + 2 \frac{A_m^2 T^2}{3} \frac{0.2}{2T} \right)$$

$$P_e = \frac{2 \cdot 6^{-2}}{6} Q\left(\frac{V_0'}{\sigma_n}\right) = \frac{5}{3} Q\left(\frac{V_0'}{\sigma_n}\right)$$